Name: ____________________________ Date: ________________ Period: _________

Parametrization Check — Key

The ellipse $\dfrac{(x+1)^2}{9} + \dfrac{(y-2)^2}{16} = 1$ is implicitly defined.

1. Identify h, k, a, and b. Write parametric equations.

$h = -1$ $\quad k = 2$ $\quad a = 3$ $\quad b = 4$

$x(t) = -1 + 3\cos t$ $\qquad y(t) = 2 + 4\sin t$

Teacher Note

A common error: students write $x(t) = h\cos t$ instead of $x(t) = h + a\cos t$ — forgetting to add the center coordinates. Prompt: "The parametrization shifts the unit ellipse to be centered at (h, k). What is the center here?"

2. Topmost and bottommost points.

$y(t) = 2 + 4\sin t$ **is maximized when** $\sin t = 1$**, i.e.,** $t = \pi/2$**.**
At $t = \pi/2$**:** $x = -1 + 3\cos(\pi/2) = -1 + 0 = -1$**,** $y = 2 + 4(1) = 6$**. Topmost point:** $(-1,\,6)$**.**
$y(t)$ **is minimized when** $\sin t = -1$**, i.e.,** $t = 3\pi/2$**.**
At $t = 3\pi/2$**:** $x = -1 + 3(0) = -1$**,** $y = 2 + 4(-1) = -2$**. Bottommost point:** $(-1,\,-2)$**.**